

SYLTRA HEALTH — Scientific Document

The scientific basis of the connected home-health intelligence layer

Version 0.1, working draft. SYLTRA ONE. For medical advisors and investment partners.

0. Why we wrote this

We want to explain, in plain language, the reasoning we built SYLTRA HEALTH on. Why we bring together what your body says and what your home sees, why we compare you to yourself rather than to an average, and why we never raise an alert before two signals agree. This is a calm review of the general evidence, not clinical research.

One boundary holds in every line: SYLTRA HEALTH is technology that helps with understanding, follow-up and early response. It does not diagnose, does not set a dose, does not predict a crisis, and does not stand in for a doctor. Read everything here inside that boundary.

The references in Section 11 are real and published; we chose them because they are authoritative and widely cited. Their fine details, volume, pages and identifiers, we verify with a scientific advisor before any external use. What you have here is an evidence framework, not a final citation set.

1. The idea, briefly

Your everyday health is not a single number; it is context. Your body's signals shift through the day, shaped by what surrounds you, the temperature, humidity, air quality, light and noise, and by what you do, your activity, sleep and hydration. Most digital tools read the body alone and leave the surroundings out, so they miss half the picture.

Put simply: when we bring your body's signals together with your home's, and compare them to what is usual for you, we get sharper context than any single signal gives. Not to diagnose, but to understand more clearly, follow up better, and respond in time.

2. Your home is part of your health

We spend most of our time indoors, and the indoor environment is tied to our comfort, sleep and ability to focus [1][2]. When carbon dioxide rises and ventilation weakens in a closed space, cognitive and decision-making scores fall measurably in controlled studies, and improve with good ventilation [3][4]. The thermal environment affects how deeply you sleep and your daily rhythm, so drifting out of a comfortable range raises night-time waking and reduces deep sleep [5]. Even light and noise shape when you sleep and when you wake [2].

None of these are captured by a band on your wrist. Placed beside your body data, they make sense of patterns that are hard to read on their own, such as sleeping better on the nights when the air was cleaner and the room was calmer.

3. We compare you to yourself, not to others

The principle we build on is that change within one person matters more than comparison to a group average. Markers like heart-rate variability differ widely between people, and even within the same person by time and state [6]. A pulse of 95 can be normal after effort and notable while you sit quietly. So we build each user a personal baseline for every metric, from a moving window, not from a generic norm for an age band.

Because an unstable baseline breeds false alarms, we announce a learning period of two to three weeks before issuing any deviation alert. We also account for your context; a weekend, travel, or an illness you told us about will suspend the deviation rules for a while. The result is that an alert based on "this is different from what is usual for you" is truer and quieter than one based on a generic threshold.

4. No alert until two signals agree

Every system that watches for change sits between two risks: missing what matters, or bothering you with a false alarm. Frequent false alarms erode trust and cause what is known as alarm fatigue, a documented safety hazard in hospitals [7][8].

So we set a clear rule: we count a change as a deviation only when at least two independent signals leave the normal range at the same time. Requiring two independent sources to agree raises the system's precision and cuts false alarms compared with a single signal, at a small, deliberate trade-off in sensitivity that suits a product built on trust.

5. When data goes missing

We hold a simple principle: missing data is not bad news. If the watch drops out or a sensor stops, we treat it as a technical fault and tell you, and we never let it enter the health-deviation calculation. Ignoring this makes every disconnection end in a false alarm, a common mistake in systems built in a hurry.

6. Medication adherence, a gap we can close

Not taking medication as prescribed is a well-known problem worldwide, with a large effect on health and cost. The World Health Organization estimated that adherence to long-term therapies in developed countries averages only about half of patients [9], with a clear effect on outcomes and cost [10]. Part of the problem is not a shortage of medicine but a matter of timing. A chronic patient buys the same medication for years, yet gaps still happen, because the run-out date is a calculated fact rather than an unknown, but no one acts on it in time.

Reminders and behavioural nudges do improve adherence, though the effect varies [11]. A precise reminder a few days before run-out, reaching the person who often cares for the patient, raises the chance of a timely refill. This is behavioural and operational value, not a pharmaceutical intervention. The system does not decide when medication is dispensed and does not change a dose; the pharmacist dispenses against a valid prescription.

7. We respond gradually, and reach the nearest with the least needed

We start from the gentlest step and do not jump, because that is what keeps your trust and lowers the noise [7][8]. First a quiet daily support, then a gentle check-in when a change is confirmed, then an alert to someone you trust if needed,

and finally an authorised escalation we do not build before formal agreements and approvals exist, so we never promise what we cannot deliver.

When we reach someone in your circle, we share only what is needed, that something warrants reassurance, without exposing your health details, unless you have authorised a specific person. And we set a weekly cap on alerts and quiet night hours, so reassurance never turns into a nuisance.

8. People of determination at the heart of the design, not an afterthought

Accessibility and human-factors principles are well established, and their essence is to present information through more than one channel, visual, audio and haptic, so it stays understandable whatever sense is available, which is the core of the accessibility content standards [12]. For us this is not a feature added at the end but a presentation layer tested with every screen from day one, because the idea is simple: one piece of information, many paths to it.

Because needs differ, we offer access profiles the user chooses at setup and can combine:

- Someone living with a motor disability runs the home by voice or a single touch, so effort is spent on what they choose, not on what the room's layout imposes.
- Someone who does not hear receives every audio cue as a light flash, a vibration and a clear notice of what happened and where, in a consistent colour and pattern that is easy to learn.
- Someone who does not see relies on precise screen reading that describes the state, high contrast and larger type, and a steady order that builds a trusted mental map of the home.
- Someone who finds speech difficult expresses themselves by touch, an on-screen choice, or ready phrases, so expression never hangs on the ability to speak, especially at a moment of need.
- Someone with higher sensory sensitivity finds a calmer environment, with gradual transitions, softer alerts, and stable surroundings set in advance to their own limits.
- Someone living with memory or focus difficulties finds gentle structure, one step at a time, a timely reminder, and few options instead of crowded menus.

In all of these, we do not draw attention to the disability; we quietly remove obstacles and leave the decision fully with the person. And one principle governs them all: greater independence, not greater surveillance, with any setting that touches sharing or contact requiring explicit consent and a clear record.

9. What we do not claim

We say it plainly: no diagnosis or condition labelling, no crisis prediction, no round-the-clock medical monitoring, no change to a dose or treatment plan, and no automatic emergency call. Regulated medical devices and deep analytics belong to the Connected Care line later, after a documented regulatory review of Software-as-a-Medical-Device status and its clinical evaluation [13][14], mindful of clinical-decision-support frameworks [15]. And everything you see, we can explain in one sentence; any alert we cannot explain, we do not send.

10. Privacy first

Your consent with us is an entity in its own right for each type of data, with a grant date and a revocation date, and revoking it takes effect at once while we show you what stopped as a result. We collect only what the experience genuinely needs, keep a clear sharing log for you and a right to deletion that reaches even derived data. And we never share your individual data with any entity that prices a service or coverage, whatever the return. Our reference for this is the controls protecting personal and sensitive-health data, which we review with regulatory counsel before any expansion.

11. References

All references below are real and published. Verify their exact details, volume, pages, identifier and edition, and update them before any external publication or peer review.

1. World Health Organization. *Guidelines for Indoor Air Quality: Selected Pollutants*. Geneva: WHO; 2010.
2. World Health Organization. *Housing and Health Guidelines*. Geneva: WHO; 2018.
3. Allen JG, MacNaughton P, Satish U, Santanam S, Vallarino J, Spengler JD. Associations of cognitive function scores with carbon dioxide, ventilation, and volatile organic compound exposures in office workers (the COGfx study). *Environmental Health Perspectives*. 2016;124(6):805–812.
4. Satish U, Mendell MJ, Shekhar K, et al. Is CO₂ an indoor pollutant? Direct effects of low-to-moderate CO₂ concentrations on human decision-making performance. *Environmental Health Perspectives*. 2012;120(12):1671–1677.
5. Okamoto-Mizuno K, Mizuno K. Effects of thermal environment on sleep and circadian rhythm. *Journal of Physiological Anthropology*. 2012;31:14.
6. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability: standards of measurement, physiological interpretation, and clinical use. *Circulation*. 1996;93(5):1043–1065.
7. The Joint Commission. *Sentinel Event Alert 50: Medical Device Alarm Safety in Hospitals*. 2013.
8. Cvach M. Monitor alarm fatigue: an integrative review. *Biomedical Instrumentation & Technology*. 2012;46(4):268–277.
9. World Health Organization (Sabaté E, ed.). *Adherence to Long-Term Therapies: Evidence for Action*. Geneva: WHO; 2003.
10. Osterberg L, Blaschke T. Adherence to medication. *New England Journal of Medicine*. 2005;353(5):487–497.
11. Nieuwlaat R, Wilczynski N, Navarro T, et al. Interventions for enhancing medication adherence. *Cochrane Database of Systematic Reviews*. 2014;(11):CD000011.
12. World Wide Web Consortium (W3C). *Web Content Accessibility Guidelines (WCAG) 2.1*. 2018.
13. IMDRF SaMD Working Group. *Software as a Medical Device (SaMD): Key Definitions*. 2013.
14. IMDRF SaMD Working Group. *Software as a Medical Device (SaMD): Clinical Evaluation*. 2017.
15. U.S. Food and Drug Administration. *Clinical Decision Support Software — Guidance for Industry and FDA Staff*. 2022.

12. How we validate, and what remains open

Our first measure is not the number of alerts nor the volume of data, but the share of alerts followed by a useful action. Alongside it we watch the ignored-alert rate, the false-alarm rate, and whether the user stays through the fourth week. Our first trial will be on the medication line, with one independent pharmacy, fifteen to twenty chronic patients, over three months, and a failure criterion written before we start and not changed after seeing the results.

A few questions remain for us to settle with the trial and with our advisor: the most suitable first data source for building the pattern baseline, how to calibrate deviation thresholds for each metric, and the ideal length of the learning period for each group.

End of document. Read it within the no-medical-claim boundary at its start. Any principle we edit here first, then in the product.